Machine Learning

Regression framework

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Regression: the stochastic framework

The stochastic/population model

- We have a quantitative response Y and p predictors $X_1, \ldots, X_p$.
- We assume there is a relationship between Y and $X = (X_1, \ldots, X_p)$

$$Y = f(X) + \epsilon,$$

where ϵ is a random error term with $\mathrm{E}(\epsilon) = 0$ and $\mathrm{Var}(\epsilon) = \sigma^2$.

- The function f is fixed but unknown and represents the systematic information that X provides about Y.

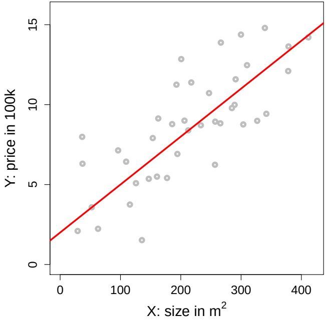
Example: [Y = house price (100k); X = size (m^2)]

$$Y = 2 + 0.03X + \epsilon$$

$$\texttt{house price} = 2 + 0.03 \times \texttt{size} + \texttt{error}$$

- This true relationship $f(X) = 2 + 0.03X$ is called the population regression line.
- Note: (X, Y) is a $(p+1)$-dimensional random vector. In this simulated example:

$$X = \mathcal{N}(0, \sigma_X^2), \text{ and } Y = 2 + 0.03X + \mathcal{N}(0, \sigma_\epsilon^2).$$



Regression: the statistical framework

The training data

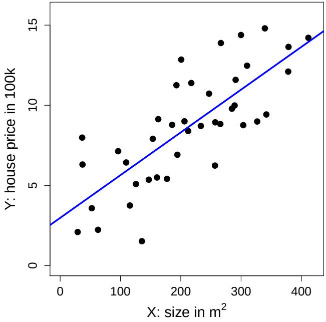
- Assume we are given n measurements $(x_1, y_1), \ldots, (x_n, y_n)$ of (X, Y), where $x_i = (x_{i1}, \ldots, x_{ip})^\top$, and
  - the inputs $x_i \in \mathbb{R}^p$ are called predictors;
  - the outputs y_i are called responses.
- We will use this data to learn the unknown relationship f between Y and X, i.e., to train (or estimate) a predictive model $\widehat{f}$.

Example: [$Y =$ house price ($100k$); $X =$ size (m^2)]

$$\texttt{house price} \approx \widehat{f}(\texttt{size}) = 2.977 + 0.027 \times \texttt{size}$$

- The training data are realizations of the random vector (X, Y), e.g.,

	X	Y
(x_1, y_1)	327	9.0
(x_2, y_2)	154	7.9
(x_3, y_3)	233	8.7
(x_4, y_4)	206	9.0
(x_5, y_5)	96	7.1
(x_6, y_6)	257	6.2
...	...	...



Why do we estimate f?

1. `Prediction`
   - A set of inputs X are readily available, but Y cannot be easily obtained.
   - We can predict Y by
   $$\widehat{Y} = \widehat{f}(X),$$
   where $\widehat{f}$ is our `predictive model` for f.
   - The function $\widehat{f}$ is often treated as `black box`: we are not concerned with the exact form of $\widehat{f}$ provided it yields good prediction for Y.

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2. `Inference`
   - We are often interested in understanding `how` Y is affected as $X_1, \dots, X_p$ change.
   - Cannot treat $\widehat{f}$ as black box, we need to know the exact form.
   - We want to answer questions such as:
     - Which predictor is associated with the response?
     - What is the relationship of the response and each predictor (positive, negative,...)?
     - Is the relationship linear or more complicated?
     - etc.

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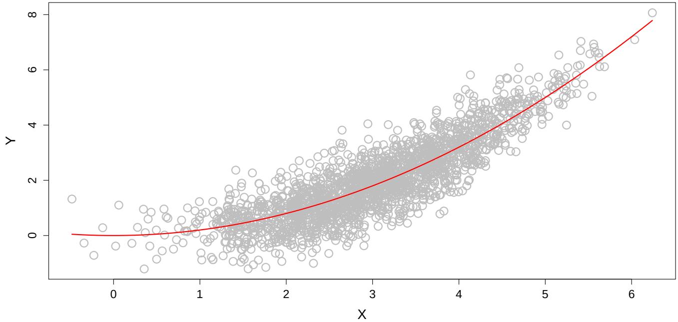
What is a good $\widehat{f}$?



- For a fixed value of X, say $X = 3$ is there an optimal prediction $\widehat{f}(X)$?
- There might be many values at $X = 3$, and a good choice, minimizing squared error, is

$$\widehat{f}(3) = \mathrm{E}(Y \mid X = 3),$$

where $\mathrm{E}(Y \mid X = 3)$ is the expected value (average) of Y given $X = 3$.

The loss function

- We have to specify what we mean with a `good prediction`. This will depend on the application.
- Introduce a `loss function` L that measures the discrepancy between a response y and its prediction $\widehat{y}$.
- In the regression setup, we typically use `squared error loss`, i.e.,

$$L(y, \widehat{y}) = (y - \widehat{y})^2.$$

- For our predictive model $\widehat{f}$, the `expected squared prediction error` is

$$\mathrm{Err}_{\widehat{f}} = \mathrm{E}[\{Y - \widehat{f}(X)\}^2],$$

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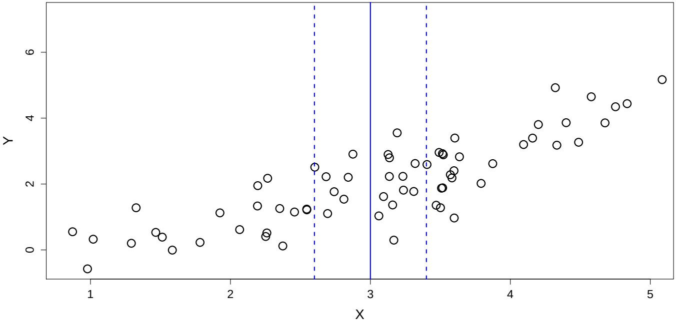
  where the expectation is taken over (X, Y).
- Ideally, we would like to find the optimal f^* that minimizes Err_{f^*}.
- In fact, f^* is the so-called regression function

$$f^*(x) = \mathrm{E}(Y|X = x),$$

  which satisfies $f^* = f$ if $Y = f(X) + \epsilon$.

How do we estimate $\widehat{f}$?



- Typically we have few or no data points with $X = 3$.
- So we cannot compute $\mathrm{E}(Y \mid X = 3)$!
- The k-Nearest-Neighbors method aims at approximating this expectation by the average

$$\widehat{f}(x_0) = \mathrm{Ave}(Y \mid X \in N_k(x_0)),$$

  where $N_k(x_0)$ are the k closest points x_i to $x_0 = 3$.
- The number k is a tuning parameter.

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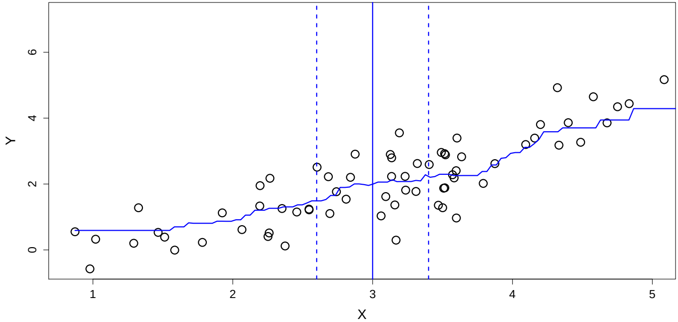

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How do we estimate $\widehat{f}$?

- Nearest neighbor averaging works good for small dimension p and a large training data size n.
- When the number of predictors $X_1, \ldots, X_p$ is larger, then kNN does not work well! The reason is the `curse of dimensionality`.

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- We will study `structured methods` that
  - need `less data` to fit;
  - may be nicely `interpretable`;
  - but at the price of `lower flexibility`.
- The difficulty is to find a good `compromise` between flexibility and parsimony.

The empirical loss: model evaluation

How can we estimate the $\mathrm{Err}_{\widehat{f}} = \mathrm{E}[\{Y - \widehat{f}(X)\}^2]$?

The test data

- For model evaluation we require a new, independent test data set $\mathcal{T}_m = \{(\tilde{x}_1, \tilde{y}_1), \ldots, (\tilde{x}_m, \tilde{y}_m)\}$.
- For each input $\tilde{x}_i$ we predict the response with our model: $\widehat{\tilde{y}}_i = \widehat{f}(\tilde{x}_i)$.
- We compare this prediction with the observed output $\tilde{y}_i$.
- The empirical prediction error of our model $\widehat{f}$ is the empirical version of $\mathrm{Err}_{\widehat{f}}$, namely

$$\mathrm{err}_{\widehat{f}} = \tfrac{1}{m} \sum_{i=1}^{m} \{\tilde{y}_i - \widehat{f}(\tilde{x}_i)\}^2,$$

also called mean squared error.

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- In practice: separate all data into training and test and do cross-validation (later).